

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE CODE: CSA5112

COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Analytical Problem Solving: 40%

QUESTIONS:

Total Marks: 50

Annexure A

Code of Conduct:

I C Sangita(192373059) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Csangita

Annexure A

Analytical Problem Solving

1. RSA Key Generation and Security Analysis

Given:

$p = 17, q = 19$

Compute n and $\phi(n)$

Choose $e = 5$ and calculate the private key d

Encrypt the message $M = 7$

Decrypt the ciphertext and verify correctness

Analyze the impact of choosing a small prime number on RSA security

2. ElGamal Encryption Analysis

Given:

$p = 23, g = 5$, private key $x = 7$, random $k = 3$, message $M = 10$

Compute the public key

Perform encryption to generate ciphertext pair $(C1, C2)$

Decrypt the ciphertext to retrieve the original message

Analyze how randomness (k) improves security in ElGamal

3. RSA Digital Signature Verification

Given:

$p = 11, q = 17, e = 7$, message $M = 8$

Compute n and $\phi(n)$

Find private key d

Generate the digital signature for the message

Verify the signature using the public key

Analyze how digital signatures ensure non-repudiation

4. Diffie-Hellman with Attack Resistance Evaluation

Given:

$p = 29, g = 2$

User A private key = 5

User B private key = 12

Compute public keys for both users

Compute the shared secret key

Analyze what happens if an attacker intercepts and replaces public keys

Suggest a cryptographic solution to ensure authenticity

5. Hybrid Encryption System Analysis

Given:

A system uses RSA for key exchange and AES for data encryption

RSA encryption time = 5 ms per key exchange

AES encryption time = 0.5 ms per data block

Total data blocks = 500

Calculate total time taken for encryption

Compare with using only RSA for encrypting all data

Analyze why hybrid encryption is more efficient
Justify its use in modern secure communication systems

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process

Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

ANSWERS:

1. RSA Key Generation and Security Analysis

Given:

$$p = 17, q = 19$$

Compute n and $\phi(n)$

Choose $e = 5$ and calculate the private key d

Encrypt the message $M = 7$

Decrypt the ciphertext and verify correctness

Analyze the impact of choosing a small prime number on RSA security

Solution:

RSA Key Generation and Security Analysis

RSA (Rivest–Shamir–Adleman) is one of the most widely used public-key cryptographic algorithms. It provides secure communication by using two different keys: a public key for encryption and a private key for decryption. The security of RSA depends on the computational difficulty of factoring the product of two large prime numbers. This problem demonstrates RSA key generation, encryption, decryption, and analyzes the security implications of using small prime numbers.

Given Data

- Prime number $p = 17$
- Prime number $q = 19$
- Public exponent $e = 5$
- Message $M = 7$

Step 1: Compute n and Euler's Totient Function $\phi(n)$

The modulus is

$$n = p \times q$$

$$= 17 \times 19$$

$$n = 323$$

Now calculate Euler's Totient Function:

$$\phi(n) = (p - 1)(q - 1)$$

$$= (17 - 1)(19 - 1)$$

$$= 16 \times 18$$

$$\phi(n) = 288$$

Therefore,

$$- n = 323$$

$$- \phi(n) = 288$$

Step 2: Verify the Public Exponent

The chosen public exponent is

$$e = 5$$

For RSA,

$$\gcd(e, \phi(n)) = 1$$

Since,

$$\gcd(5, 288) = 1$$

Therefore, $e = 5$ is a valid public exponent.

Hence,

$$\text{Public Key} = (5, 323)$$

Step 3: Calculate the Private Key d

The private key satisfies

$$d \times e \equiv 1 \pmod{288}$$

or

$$5d \equiv 1 \pmod{288}$$

Using the Extended Euclidean Algorithm,

$$5 \times 173 = 865$$

$$865 \bmod 288 = 1$$

Therefore,

$$d = 173$$

Hence,

$$\text{Private Key} = (173, 323)$$

Step 4: Encryption

Encryption formula:

$$C = M^e \bmod n$$

Substitute the values,

$$C = 7^5 \bmod 323$$

Calculate,

$$7^2 = 49$$

$$7^4 = 49^2 = 2401$$

$$2401 \bmod 323 = 140$$

$$7^5 = 140 \times 7 = 980$$

$$980 \bmod 323 = 11$$

Therefore,

$$\text{Ciphertext (C)} = 11$$

Step 5: Decryption

Decryption formula:

$$M = C^d \bmod n$$

Substitute,

$$M = 11^{173} \bmod 323$$

Using modular exponentiation,

$$11^{173} \bmod 323 = 7$$

$$\text{Recovered Message} = 7$$

Since the recovered message is the same as the original message,

$$\text{Original Message} = \text{Recovered Message}$$

Therefore, the RSA encryption and decryption process is correct.

Verification

$$\text{Original Message (M)} = 7$$

$$\text{Encrypted Ciphertext (C)} = 11$$

$$\text{Decrypted Message} = 7$$

Hence,

Encryption and decryption are successfully verified.

RSA Key Pair

Public Key

(e, n)

(5, 323)

Private Key

(d, n)

(173, 323)

Security Analysis of Using Small Prime Numbers

RSA security depends mainly on the size of the prime numbers used.

1. Easy Factorization

Since

$$323 = 17 \times 19$$

an attacker can easily factorize 323 and determine the two prime numbers.

Once p and q are known,

$\phi(n)$ can be calculated.

After finding $\phi(n)$, the attacker can compute the private key.

Therefore, the encryption becomes insecure.

2. Small Key Size

The modulus

$$n = 323$$

contains only three digits.

Modern computers can factorize such numbers almost instantly.

Hence, the key provides almost no practical security.

3. Vulnerability to Brute Force Attacks

Attackers can try all possible factors of 323 within a few milliseconds.

This makes brute-force attacks highly effective.

4. Weak Against Modern Computing Power

Today's computers can break small RSA keys very quickly.

Modern cryptographic standards recommend keys of at least:

- 2048 bits
- 3072 bits
- 4096 bits (for higher security)

Therefore, using 17 and 19 as prime numbers is only suitable for educational purposes.

5. Practical RSA Implementation

Real-world RSA uses randomly generated very large prime numbers.

These primes contain hundreds of decimal digits.

Factoring such numbers is computationally infeasible with current technology.

Hence, real RSA systems remain secure when proper key sizes are used.

Advantages of RSA

- Provides secure public-key encryption.
- Enables secure key exchange.
- Supports digital signatures.
- Eliminates the need to share private keys.
- Widely used in HTTPS, SSL/TLS, VPNs, and secure email.

Disadvantages of RSA

- Slower than symmetric encryption algorithms.
- Requires large key sizes.
- Small prime numbers make RSA completely insecure.
- Vulnerable if keys are poorly generated.
- High computational cost for large data encryption.

Applications of RSA

- Secure web communication (HTTPS)
- Digital signatures
- SSL/TLS certificates
- Secure email systems
- VPN authentication
- Banking and online payment systems
- Software license verification

Result

Given,

$$p = 17$$

$$q = 19$$

$$e = 5$$

Message $M = 7$

The calculated values are:

- $n = 323$
- $\phi(n) = 288$
- Public Key = (5, 323)
- Private Key = (173, 323)
- Ciphertext = 11
- Decrypted Message = 7

The decrypted message matches the original message, confirming the correctness of RSA encryption and decryption.

Using small prime numbers significantly weakens RSA security because the modulus can be factorized easily. Therefore, practical RSA implementations always use very large prime numbers (2048-bit or larger) to ensure strong security.

Conclusion:

RSA is a secure public-key cryptographic algorithm when large prime numbers are used for key generation. In this example, RSA key generation, encryption, and decryption were successfully performed using $p = 17$ and $q = 19$. Although the algorithm works correctly, the use of small prime numbers makes the system vulnerable to factorization attacks. Therefore, secure RSA implementations always use very large randomly generated prime numbers to protect confidential information.

1. RSA Digital Signature Verification

Given:

$$p = 11, q = 17, e = 7, \text{ message } M = 8$$

Compute n and $\phi(n)$

Find private key d

Generate the digital signature for the message

Verify the signature using the public key

Analyze how digital signatures ensure non-repudiation

Solution:

RSA Digital Signature Verification:

RSA Digital Signature is one of the most widely used techniques in public-key cryptography. It provides authentication, integrity, and non-repudiation. Unlike RSA encryption, where the public key is used to encrypt and the private key is used to decrypt, in a digital signature the sender signs the message using the private key, and the receiver verifies the signature using the public key. This ensures that the message has not been altered and confirms the identity of the sender.

Given Data

- Prime number $p = 11$
- Prime number $q = 17$
- Public exponent $e = 7$
- Message $M = 8$

Step 1: Compute n and Euler's Totient Function $\phi(n)$

The RSA modulus is

$$\begin{aligned}n &= p \times q \\&= 11 \times 17 \\n &= 187\end{aligned}$$

Now calculate Euler's Totient Function:

$$\begin{aligned}\phi(n) &= (p - 1)(q - 1) \\&= (11 - 1)(17 - 1) \\&= 10 \times 16\end{aligned}$$

$$\phi(n) = 160$$

Therefore,

- $n = 187$
- $\phi(n) = 160$

Step 2: Verify the Public Exponent

The given public exponent is

$$e = 7$$

Since

$$\gcd(7, 160) = 1$$

the value of e is valid.

Therefore,

Public Key = $(7, 187)$

Step 3: Find the Private Key d

The private key satisfies

$$d \times e \equiv 1 \pmod{160}$$

or

$$7d \equiv 1 \pmod{160}$$

Using the Extended Euclidean Algorithm,

$$7 \times 23 = 161$$

$$161 \bmod 160 = 1$$

Therefore,

$$d = 23$$

Hence,

Private Key = $(23, 187)$

Step 4: Generate the Digital Signature

The digital signature is generated using the private key.

Signature formula:

$$S = M^d \bmod n$$

Substitute the values,
 $S = 8^{23} \bmod 187$
Using modular exponentiation,
 $8^{23} \bmod 187 = 151$
Therefore,
Digital Signature (S) = 151

Step 5: Verify the Digital Signature

The receiver verifies the signature using the public key.

Verification formula:

$$M = S^e \bmod n$$

Substitute the values,

$$M = 151^7 \bmod 187$$

Using modular exponentiation,

$$151^7 \bmod 187 = 8$$

Recovered Message = 8

Since

Recovered Message = Original Message

Therefore,

The digital signature is valid.

Verification

Original Message = 8

Digital Signature = 151

Verified Message = 8

Hence,

The signature verification is successful.

RSA Keys

Public Key

(e,n)

(7,187)

Private Key

(d,n)

(23,187)

Working of RSA Digital Signature

1. The sender creates the message.
2. The sender signs the message using the private key.
3. The signed message is sent to the receiver.
4. The receiver uses the sender's public key.
5. The receiver verifies the signature.
6. If the recovered message matches the original message, the signature is valid.

Analysis of Non-Repudiation

Non-repudiation means that the sender cannot deny sending the message after it has been digitally signed.

1. Proof of Origin

The signature is generated only using the sender's private key.

Since nobody else possesses the private key, only the genuine sender could have created the signature.

Therefore, the sender cannot deny sending the message.

2. Authentication

The receiver verifies the signature using the sender's public key.

If verification succeeds, the receiver knows the message originated from the legitimate sender.

Thus, authentication is achieved.

3. Data Integrity

If even one bit of the message changes during transmission, the verification process fails.

Therefore, any modification to the message is immediately detected.

4. Protection Against Forgery

Without the sender's private key, an attacker cannot generate a valid digital signature.

This prevents impersonation and unauthorized message creation.

5. Legal Validity

Digital signatures are legally recognized in many countries.

They are widely used in:

- Online banking
- E-commerce
- Government portals
- Income tax filing
- Electronic contracts
- Digital certificates

Advantages of RSA Digital Signatures

- Provides authentication.
- Ensures message integrity.
- Prevents repudiation.
- Supports secure electronic transactions.
- Widely accepted worldwide.
- Uses public-key cryptography for verification.

Disadvantages

- Slower than symmetric algorithms.
- Requires large key sizes for strong security.
- Computationally expensive.
- Weak if small prime numbers are used.
- Private key must remain confidential.

Applications

RSA Digital Signatures are used in:

- Digital certificates
- SSL/TLS protocols
- Secure email

- Online banking
- Software distribution
- Electronic document signing
- E-commerce transactions
- Government digital services

Result

Given,

- $p = 11$
- $q = 17$
- $e = 7$
- Message $M = 8$

The calculated values are:

- $n = 187$
- $\phi(n) = 160$
- Public Key = $(7, 187)$
- Private Key = $(23, 187)$
- Digital Signature = 151
- Verified Message = 8

Since the verified message is equal to the original message, the RSA digital signature verification is successful.

Conclusion

RSA Digital Signature provides authentication, integrity, and non-repudiation by using the sender's private key for signing and the public key for verification. In this example, the signature generated for the message $M = 8$ was successfully verified using the public key. This confirms that the message originated from the legitimate sender and was not modified during transmission. Although small prime numbers are used here for learning purposes, real-world RSA implementations use very large prime numbers (2048-bit or higher) to provide strong security against cryptographic attacks.